

July 2026

Alireza Nasirzadeh, PhD

Research Professor · Assistive and Rehabilitation Robotics Laboratory

✉ alireza@cau.ac.kr · 🌐 [Website](#)

PROFESSIONAL EXPERIENCE

Research Professor

Feb 2022 – Present

*Assistive and Rehabilitation Robotics Laboratory, School of Mechanical Engineering
Chung-Ang University, South Korea*

Biomechanist

Jan 2020 – Oct 2021

Athlete Evaluation Department, I.R.I. National Paralympic Committee, Iran

Assistant Professor

Sep 2019 – Nov 2021

Faculty of Physical Education and Sport Sciences, Kharazmi University, Iran

Teaching Assistant

Jan 2016 – May 2019

Various Universities, Iran

EDUCATION

PhD in Sports Biomechanics

Sep 2012 – Sep 2016

Kharazmi University, Iran

Thesis: Comparison of Selected Kinematical, Kinetical, and Functional Lower Limb Variables during Walking and Running at Different Speeds among Young Male Athletes Focusing on Limb Dominance

MSc in Physical Education and Sports Sciences

Sep 2009 – May 2011

University of Birjand, Iran

Dissertation: Architectural Characteristics of the Upper Extremity Skeletal Muscles in Young Male Table Tennis Players

BSc in Mathematics

Sep 2003 – May 2009

University of Birjand, Iran

RESEARCH AREAS

Movement Biomechanics · Wearable Robotics & Exosuits · Musculoskeletal Modeling · Dynamic Stability and Balance · Rehabilitation Engineering · Injury Mechanics

PUBLICATIONS

Nasirzadeh, A. et al. (2026). [Biomechanical Effects of Passive Exosuit Assistance on Tibiofemoral Loading and Dynamic Stability during Downhill Walking](#). *Journal of NeuroEngineering and Rehabilitation*.

Park, C., **Nasirzadeh, A.**, Lee, K., Kim, G., & Lee, G. (2026). [Real-Time Estimation of User Adaptation During Hip Exosuit-Assisted Walking Using Wearable Inertial Measurement Unit Data and Long Short-Term Memory Modeling](#). *Biomimetics*, 11(2), 96.

Nasirzadeh, A. et al. (2025). [Harnessing Resistance: The Impact of a Passive Exosuit on Golf Swing](#)

[Biomechanics](#). *Journal of Biomechanics*, 192, 112958.

Choe, E., Moon, J., Ryu, J., Yang, S., **Nasirzadeh, A.**, Kong, S., Choi, Y., & Lee, G. (2025). [Low-Profile, Shoe-Type Ankle-Foot Orthosis with Active Variable Ankle Stiffness via Wire-Fabric Compression Mechanism](#). *Biomimetics*, 10(8), 539.

Kim, Y., Park, S., Yang, S., **Nasirzadeh, A.**, Lee, G. (2024). [The Interplay Between Muscular Activity and Pattern Recognition of Electro-Stimulated Haptic Cues During Normal Walking: A Pilot Study](#). *Bioengineering*, 11, 1248.

Nasirzadeh, A. et al. (2023). [The Effects of a Custom-Designed High-Collar Shoe on Muscular Activity, Dynamic Stability, and Leg Stiffness: A Biomimetic Approach Study](#). *Biomimetics*, 8, 274.

Park, J., Nam, K., Yun, J., Moon, J., Ryu, J., Park, S., **Nasirzadeh, A.** et al. (2023). [Effect of hip abduction assistance on metabolic cost and balance during human walking](#). *Science Robotics*, 8(83), eade0876.

Nasirzadeh, A. et al. (2023). [Influence of circumferential ankle pressure of shoe collar on the kinematics, dynamic stability, electromyography, and plantar pressure during normal walking](#). *PLOS ONE*, 18(2), e0281684.

Nasirzadeh, A. et al. (2017). [A review of selected factors that affect gait symmetry](#). *Physical Treatments*, 7(1), 3-12.

Nasirzadeh, A. et al. (2015). [Multivariate analysis of 200-m front crawl swimming performance in young male swimmers](#). *Acta of Bioengineering and Biomechanics*, 17(3), 137-43.

Nasirzadeh, A. et al. (2014). [Selected anthropometrical, muscular architecture, and biomechanical variables as predictors of 50-m performance of front crawl swimming in young male swimmers](#). *Science and Sports*, 29(5), e75-e81.

Nasirzadeh, A. et al. (2014). [Relationship between sprint performance of front crawl swimming and muscle fascicle length in young swimmers](#). *Journal of Sports Science and Medicine*, 13(3), 550-556.

PRESENTATIONS

PODIUM PRESENTATIONS

Mediolateral Balance Recovery during Walking using H-Medi Exosuit (2026). **44th International Society of Biomechanics in Sports Conference (ISBS)**, Loughborough, UK.

Design and Validation of a Passive Exosuit to Enhance Golf Swing Performance (2025). **7th International Conference on Medicine in Novel Technology and Devices**, Hangzhou, China.

[Power Up the Swing: A Biomechanical Study Towards a Golf Exosuit](#) (2024). **42nd International Society of Biomechanics in Sports Conference (ISBS)**, Salzburg, Austria.

[Active Variable Compression-Shoes: Design and Evaluation of an External Ankle Supporter to Prevent Ankle Sprain via Active Compression Mechanism](#) (2024). **10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob)**, Heidelberg, Germany.

[The effect of gender and anthropometrical parameters on the dynamics of sprint start among young elite sprinters](#) (2012). **XXX International Society of Biomechanics in Sports (ISBS) Conference**, Melbourne, Australia.

POSTER PRESENTATIONS

Reducing Tibiofemoral Joint Force During Downhill Walking Using Various Passive Exosuits: A Pilot Study (2025). **XXX Congress of the International Society of Biomechanics (ISB)**, Stockholm, Sweden.

Passive Collar Stiffness of a Custom-designed High-Collar Shoe (2023). **XXIX Congress of the International Society of Biomechanics (ISB)**, Fukuoka, Japan.

AWARDS & HONORS

Outstanding Student Department of Sports Biomechanics, Kharazmi University	2016
Outstanding Presentation Award in Biomechanics First National Conference on New Findings in Sport Sciences	2015
New Investigator Award Sixth National Conference on Physical Education and Sport Sciences	2011

SERVICE & OUTREACH

Ad-Hoc Reviewer

Journal of Sports Sciences

Sports Biomechanics

International Journal of Athletic Therapy & Training

Physical Treatments

PLOS ONE

IEEE Transactions on Neural Systems & Rehabilitation Engineering

MEMBERSHIPS & AFFILIATIONS

International Society of Biomechanics

International Society of Biomechanics in Sports

REFERENCES

[Giuk Lee](#)

Associate Professor, Department of Mechanical Engineering, Chung-Ang University

✉ giuklee@cau.ac.kr

[Yong Ung Kwon](#)

Associate Professor, College of Sport Sciences, Chung-Ang University

✉ ykwon@cau.ac.kr

[Saeed Ilbeigi](#)

Associate Professor, Physical Education and Sports Sciences Department, University of Birjand

✉ silbeigi@birjand.ac.ir